

Geometry
Unit 13
Lesson 9 Practice

Name _____
Date _____
Period _____

Najira is using a computer software to design the layout of her bathroom remodel. The base of the bathroom creates $DEFG$ with coordinates $D(7, -2)$, $E(1, -6)$, $F(3, -9)$, and $G(9, -5)$.

1. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DE}$	
$\overline{EF}$	
$\overline{FG}$	
$\overline{DG}$	

2. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DF}$	
$\overline{EG}$	

3. Finish the statement.

$DEFG$ is classified as a _____ because ...

4. If 1 unit on the coordinate plane equals 3 feet (ft), what are the dimensions of the base of the bathroom? Round to the nearest thousandth.

5. $PRNT$ has vertices at $P(3, -8)$, $R(2, 0)$, $N(-5, 4)$, and $T(-4, -4)$.

$PRNT$ is classified as a(n) _____ because ...

Ginny is designing a template for her new 3D printer. She created two quadrilaterals using her software.

<i>MSFC</i>	<i>TDPR</i>
$M(0, -7), S(4, -3), F(6, 5), C(2, 1)$	$T(5, 7), D(7, 4), P(0, 4), R(2, 7)$

6. Which quadrilateral can be classified as a trapezoid? Show your work.
7. List the properties of a trapezoid that you used.
8. If 1 unit of the coordinate plane equals 4 inches (in.), what are the measurements of the sides of the trapezoid?

Fintan was graphing square $RD\overline{CP}$, but mixed up his vertices. He knows that point R has coordinates $(4, -5)$ and point P has coordinates $(2, -2)$. His two options for points C and D are shown.

Point C	Point D
$(-1, -4)$	$(-1, -7)$
$(-1, -3)$	$(1, -7)$

9. Which coordinates are correct for points C and D ? Justify your answer or show your work.
10. Which properties of a square did you use to determine the correct coordinates for points C and D ?